4 **(a)** **(i)** The electric field strength E at a point is the ***electric force per unit positive charge*** acting on a small test charge placed at the point. 1

(ii) The work done by a force on an object is the product of the force and the displacement in the direction of the force, work done = $F\,d$. 1

From part (i), the electrical force is $F_E = q\,E$ (to the right). The charge is moved to the left. Hence, work done by F_E = $\boldsymbol{-\,q\,E\,d}$. (negative sign necessary)

(iii) Work done by external force = gain in electrical potential energy 1
Hence, gain in electrical potential energy = - work done by electrical force
$q\,(V_Y - V_X) = q\,E\,d$
$V = V_Y - V_X = \boldsymbol{E\,d}$

(iv) Electric field lines are always perpendicular to equipotential lines. Hence, students should draw **a vertical line passing through point X**. 1

(v) Since electric fields point from higher to lower potential, **point Y** is at a higher potential (than point X). 1

(b) **(i)** By the principle of conservation of energy,
loss in potential electrical energy = gain in kinetic energy.
Since the electrons started from rest,
$q\,V = \frac{1}{2}\,m\,v_{final}^2$
$(1.60 \times 10^{-19})\,(10 \times 10^3) = \frac{1}{2}\,(9.11 \times 10^{-31})\,v_{final}^2$ 1
$v_{final} = 5.93 \times 10^7\ \text{m s}^{-1}$ 1

(ii) The most energetic photon is produced when 100% of the kinetic energy of the incident electron goes into producing it.
$q\,V = h\,c\,/\,\lambda$
$(1.60 \times 10^{-19})\,(10 \times 10^3) = (6.63 \times 10^{-34})\,(3.00 \times 10^8)\,/\,\lambda$ 1
$\lambda = 1.24 \times 10^{-10}$ m 1

[Total: 9]

5 **(a)** When the resistance of the variable resistor is increased, it results in a decrease in